Non-Linear Joint Models for Clinical Trials

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Joint Models

A joint model simultaneously models longitudinal (time-dependent) data and Time-To-Event (TTE) data whilst accounting for censoring [1].

Consider an example of cancer patients undergoing a specific treatment. In this case, the longitudinal data could be the size of a tumor and the event of interest could be whether the patient dies under that treatment.

Longitudinal Model

A longitudinal model, say $m(t, \underline{\Psi}_i)$, allows us to model a variable of interest (often referred to as the biomarker) as time progresses [1].

In our example, modeling how a patients tumor changes in size allows predictions to be made regarding patient outcomes so that necessary decisions can be made in time.

Different patients will, of course, exhibit differences in how their tumors progress and how they respond to any treatments. Thus, the longitudinal model is typically a **mixed-effect model**, utilizing the fixed and random effects of patient i ($\underline{\Psi}_i$) to account for within-patient correlation and allowing for the modelling of multiple patients.

Survival Model

TTE data is modelled through a survival model. This is done by determining the relevant **hazard function** $h_i(t)$, defined as the risk of a patient experiencing the event of interest at a specific time. i.e.

$$h_i(t) = \lim_{\Delta t \to 0} \frac{\mathrm{P}(t < T_i \leq t + \Delta t \mid T_i > t)}{\Delta t}$$

where T_i is the time of event for patient i [1].

It is within the survival model that censoring is accounted for. Censoring is defined as the case where the time of event is not observed in a patient for any reason. For example, a patient may no longer be able to participate in the study after moving or the event may simply have not occurred during the study period. **In survival models, censoring is assumed random.**

Why a Joint Model?

We expect that a patient with a large tumour to have a lower survivability. Additionally, once those with low survivability die, we expect that the remaining patients to have smaller tumours, biasing measurements. Thus, **the longitudinal model and survival model both effect each other and are not independent**. Therefore, **a joint model** that models both processes and their dependence on each other **jointly** is required.

This allows joint models to account for non-random censoring. For example, a patient may be dissatisfied with the treatment results under the study and decide to pull out to seek another treatment.

Ultimately, by making use of a joint model, one can create a more accurate model than by modelling either process alone, allowing for more accurate conclusions about the efficacy of the treatment being studied.

Non-Linear Joint Models

A typical, linear joint model may be:

$$h_i(t, \underline{\Psi}_i) = h_0(t) \cdot e^{\beta \cdot f(t, \underline{\Psi}_i) + \gamma \cdot g(\underline{\Psi}_i)}$$

where h_0 is the baseline hazard function, $f(t, \underline{\Psi}_i)$ is the indirect effects and is some function of $m(t, \underline{\Psi}_i)$, and $g(\underline{\Psi}_i)$ is the direct effects.

However, say that instead of simply $\beta \cdot f(t, \underline{\Psi}_i)$, we had some **non-linear** function of f, such as a square root function, a spline, or a log function. This would then be a **non-linear joint model**.

Application

Primary Biliary Cirrhosis (PBC) data gathered by the Mayo Clinic as present in the JMbayes2 package is analysed in R [2] [3] [4]. Eight variables are selected:

- serum bilirubin concentration
- SGOT concentration
- albumin concentration
- presence of an edema
- platelet count
- alkaline phosphatase concentrations
- ascites
- presence of spider-like patterns on the skin

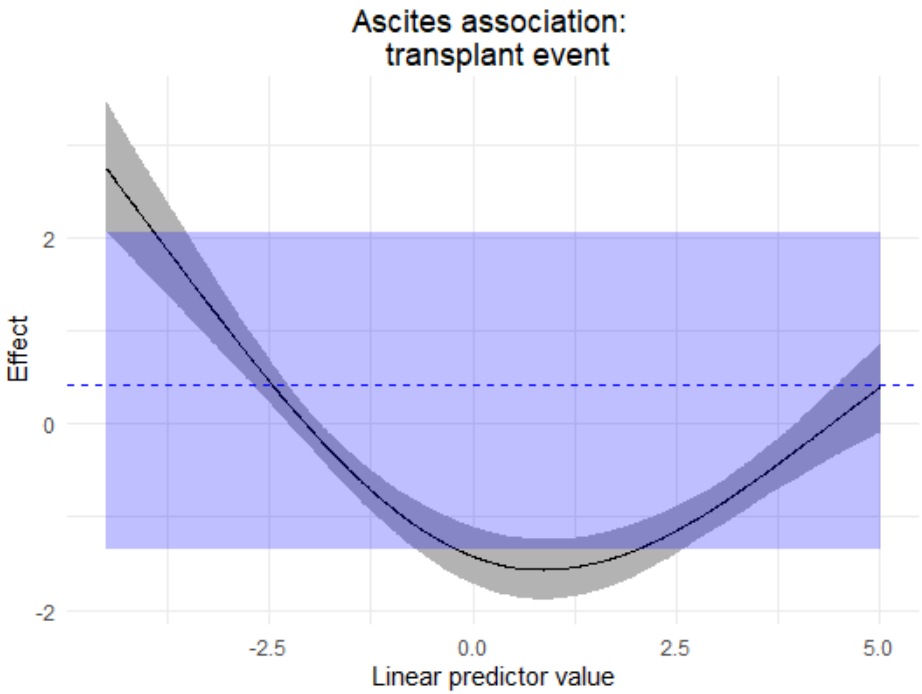
For each variable linear and non-linear joint models are fitted for both possible events. One event is death and the other is liver transplant. BIC and WAIC values [6] are determined for all models.

Results

		Linear model		Non-linear model	
Variable	Event	DIC	WAIC	DIC	WAIC
Serum bilirubin	Death	4902.609	4946.742	**4902.13**	**4944.485**
	Transplant	4071.672	4117.198	**4070.28**	**4115.777**
SGOT	Death	19497.23	20037.12	**19482.86**	**20022.05**
	Transplant	18544.92	19069.8	**18534.57**	**19061.93**
Albumin	Death	**2764.856**	**3014.882**	2766.197	3048.331
	Transplant	1908.689	2069.652	**1895.431**	**2007.982**
Edema	Death	**2582.435**	**2530.852**	2604.887	2576.556
	Transplant	1730.421	**1664.705**	**1708.107**	1677.515
Platelets	Death	21196.92	21323.67	**21191.45**	**21315.79**
	Transplant	**20220.62**	**20349.74**	20224.22	20352.82
Alkaline	Death	29024.2	29178.92	**29018.79**	**29173.92**
	Transplant	28015.65	28169.95	**27997.13**	**28155.36**
Ascites	Death	**2185.548**	**2159.431**	2226.738	2231.49
	Transplant	1369.197	1343.332	**1278.394**	**1259.9**
Spiders	Death	2868.248	2807.265	**2771.853**	**2711.456**
	Transplant	1929.686	1858.482	**1922.981**	**1851.726**

Table 1: DIC and WBIC for fitted models (better values bolded)

Non-linear joint models had better DIC in $\frac{12}{16}$ (75%) of cases and a better WAIC in $\frac{11}{16}$ (69%) of cases.



Linear joint model:

- Wide confidence interval
- Not significantly different from zero

Non-linear joint model:

- Narrow confidence interval
- Significantly different from zero
- Higher risk at extreme values
- Lower risk at moderate values
- Better DIC and WAIC

Two very different conclusions!

Conclusion

In the **majority** of cases, **non-linear** joint models had **better DIC and WAIC**, drew **more meaningful** conclusions, and **captured complex processes** that linear joint models could not. This shows that linear joint models are often insufficient within a clinical context and that **relying solely on linear joint models can lead to errors in patient care and treatment.**

Further studies could look at a wide variety of datasets, expand analysis to multivariate joint models, or investigate differences in predictive ability of linear and non-linear joint models.


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